# Deep Reinforcement Learning

Reinforcement Learning

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- Value-based deep RL: Deep-Q Networks, stabilization principles
- Policy gradient methods: direct policy search, policy gradient theorem, REINFORCE, baselines
- Actor-critic methods: Advantage Actor-Critic, Proximal Policy Optimization, Soft Actor-Critic

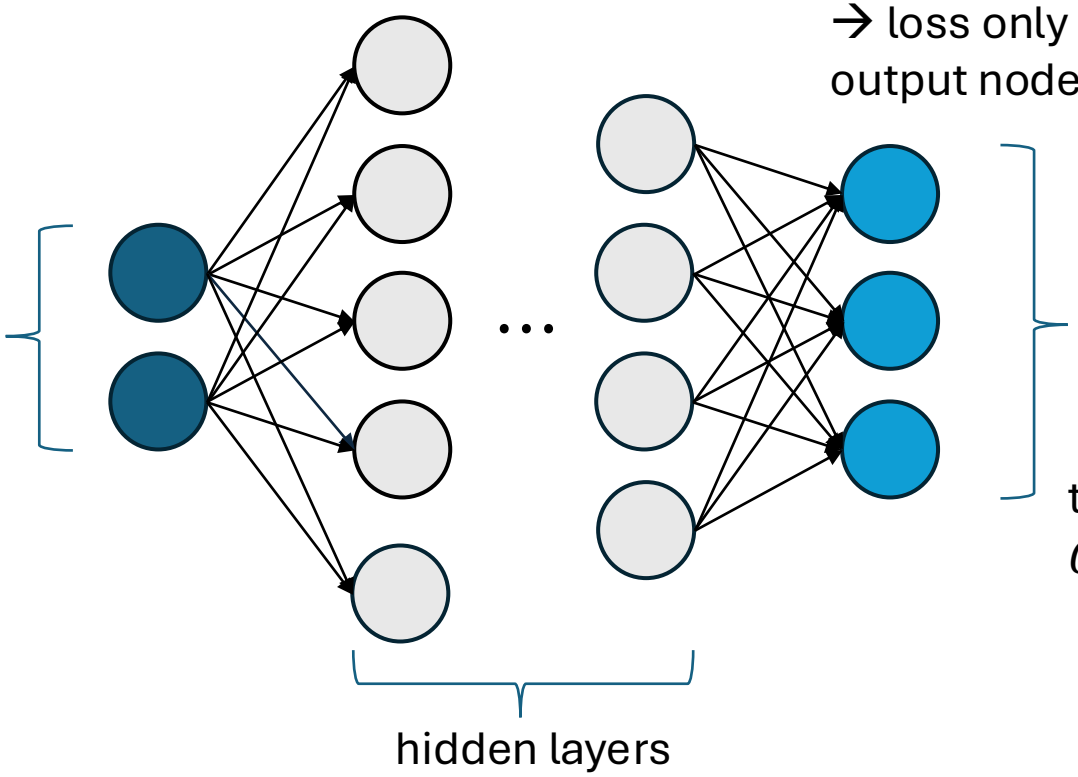
# Value-Based Deep RL

# Deep Q-Network (DQN)

idea: approximation of tabular $Q$-function by neural network

only one action for each training observation
→ loss only calculated from the corresponding output node (other outputs calculated but ignored)

values describing the state




$Q$-values for the different actions ( → discrete actions)

target:
$$Q(s_t, a_t) = r_{t+1} + \gamma \max_a Q(s_{t+1}, a_{t+1})$$

# Distribution Shift in RL

the “deadly triad”:

bootstrapping + off-policy + function approximation

e.g., Q-learning

e.g., (deep) neural network

the data the network sees is nonstationary, leading to instability

# Stabilization Principles

general principles for stabilizing value-based deep RL:

- replay buffers
- target networks
- clipping
- double Q
- prioritized replay

# Key Components of DQN

two important tricks for training stabilization

experience replay:

training samples (or mini batches) drawn at random from previously stored experiences → remove correlations in observation sequence

separate target network:

copy of $Q$-network, but only periodically updated → reduce target correlations due to bootstrapping

# Continuous Actions

instead of one output node for each discrete action, need to feed continuous action as input into the network

→ one output node for Q-value of the given action

problem: which action to choose then at runtime?

→ would need to run all possible actions through the network to find best Q-value

better: instead of Q-value, output action (i.e., policy) directly

# Policy Gradient Methods

# Direct Policy Search

goal: find an action policy without using value functions

parameterized policy $\pi(a_t|s_t;\boldsymbol{w})$:

probability to take different actions (target) given a state (variables/features) and parameters (e.g., neural network weights)

→ search for good parameters directly in the policy space using optimization algorithms

# Objective

goal of policy search: maximizing expected return

→ objective function for parameterized policy $\pi(a_t|s_t;\boldsymbol{w})$:

$$J(\boldsymbol{w}) = \mathbb{E}_{\tau\sim\pi}[R(\tau)] = \int p_{\boldsymbol{w}}(\tau)\, R(\tau)\, d\tau$$

with trajectory $\tau = (s_0, a_0, s_1, a_1, \cdots)$,
return $R(\tau) = \sum_t \gamma^t r_t$,
and trajectory probability $p_{\boldsymbol{w}}(\tau) = \rho_0(s_0)\ \prod_t \pi(a_t|s_t;\boldsymbol{w})\ P(s_{t+1}|s_t, a_t)$

initial state distribution

# Log-Likelihood Trick

using

$$\nabla_{\boldsymbol{w}}\, p_{\boldsymbol{w}}(\tau) = p_{\boldsymbol{w}}(\tau)\, \nabla_{\boldsymbol{w}} \log p_{\boldsymbol{w}}(\tau)$$

$$\nabla_{\boldsymbol{w}}\, J(\boldsymbol{w}) = \int \nabla_{\boldsymbol{w}}\, p_{\boldsymbol{w}}(\tau)\, R(\tau)\, d\tau = \int p_{\boldsymbol{w}}(\tau)\, \nabla_{\boldsymbol{w}} \log p_{\boldsymbol{w}}(\tau)\, R(\tau)\, d\tau$$

$$= \mathbb{E}_{\tau\sim\pi}[\underbrace{\nabla_{\boldsymbol{w}} \log p_{\boldsymbol{w}}(\tau)}\, R(\tau)] = \mathbb{E}_{\tau\sim\pi}\left[\left(\sum_t \nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})\right) R(\tau)\right]$$

$$\boxed{\begin{gathered}\log p_{\boldsymbol{w}}(\tau) = \log \rho_0(s_0) + \sum_t \log \pi(a_t|s_t;\boldsymbol{w}) + \sum_t \log P(s_{t+1}|s_t, a_t) \\ \Rightarrow \nabla_{\boldsymbol{w}} \log p_{\boldsymbol{w}}(\tau) = \sum_t \nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})\end{gathered}}$$

# Policy Gradient Theorem

replace the full return $R(\tau)$ of a trajectory $\tau$ with the future return $G_t = \sum_{t'=t+1}^{\infty} \gamma^{t'-t} r_{t'}$ for each time step $t$:

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \mathbb{E}_{\pi}\left[\sum_{t} \nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})\, G_t\right]$$

- $R(\tau) = G_t + \overbrace{(\text{past rewards until } t)}^{\text{independent of current action}}$
- $\mathbb{E}_{a\sim\pi}[\nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})] = 0$

→ past terms vanishes

→ gradient of an agent's performance with respect to its policy parameters, enabling policy optimization through gradient ascent

# Side Note: Score Function Property

$$\mathbb{E}_{a_t \sim \pi}[\nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})]$$
$$= \sum_a \pi(a_t|s_t;\boldsymbol{w}) \nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w}) = \sum_a \nabla_{\boldsymbol{w}} \pi(a_t|s_t;\boldsymbol{w}) = 0$$

$$\nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w}) = \frac{\nabla_{\boldsymbol{w}} \pi(a_t|s_t;\boldsymbol{w})}{\pi(a_t|s_t;\boldsymbol{w})}$$

$$\sum_a \pi(a_t|s_t;\boldsymbol{w}) = 1$$
$$\Rightarrow \nabla_{\boldsymbol{w}} \sum_a \pi(a_t|s_t;\boldsymbol{w}) = 0$$
$$\Rightarrow \sum_a \nabla_{\boldsymbol{w}} \pi(a_t|s_t;\boldsymbol{w}) = 0$$

# REINFORCE

approximate expectation in policy gradient theorem using MC samples:

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) \approx \frac{1}{n} \sum_{i=1}^{n} \sum_{t} \nabla_{\boldsymbol{w}} \log \pi \left( a_t^{(i)} | s_t^{(i)}; \boldsymbol{w} \right) G_t^{(i)}$$

batch of full episodes

"weighting" with observed (discounted) return

$$\boldsymbol{w} \leftarrow \boldsymbol{w} + \eta \cdot \nabla_{\boldsymbol{w}} J(\boldsymbol{w})$$

update after each episode (SGD, $n = 1$)
or after several ones (mini-batch gradient descent)

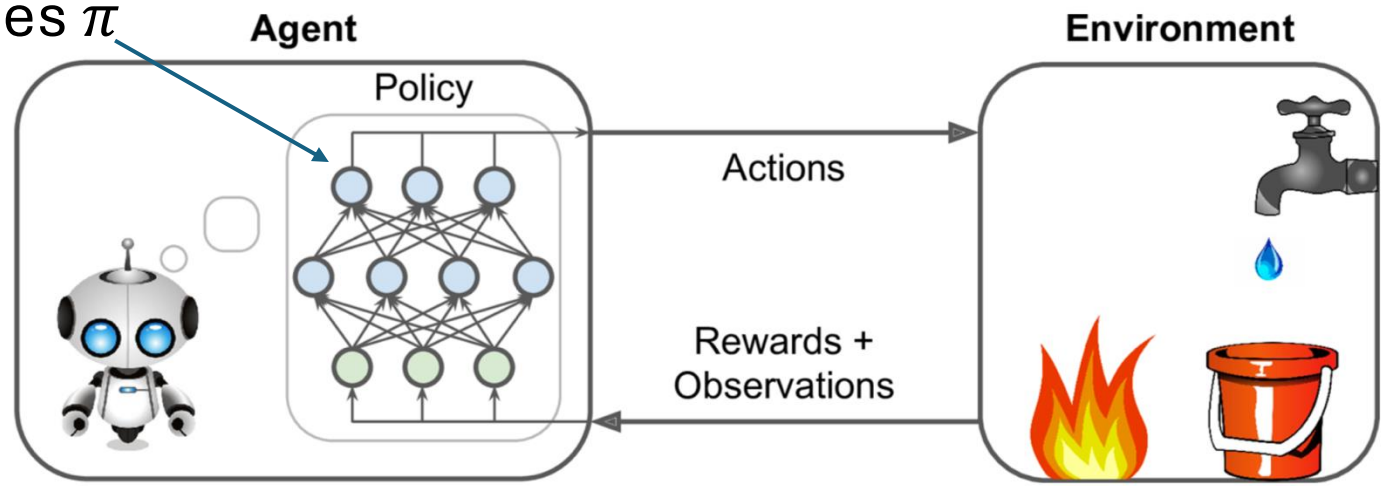
typical implementation: policy gradients → neural network gradients

# Implementation of Policy Gradient Method

neural network with actions as output/target: policy network

action probabilities $\pi$

$\log \pi$: score function




observed returns
(MC sampling)

$$\boxed{\text{loss: } -\sum_t \log \pi(a_t|s_t;\boldsymbol{w})\, G_t}$$

gradient ascent
→ actions with higher rewards

choose action according to probability
→ implicit exploration, on-policy method

# Continuous Actions

one of the main motivations to go from value-based to policy gradient methods was to deal with continuous actions

but so far only considered discrete actions → need to change network output layer to deal with continuous ones

network output:

- discrete actions: one probability (i.e., one output node) for each possible action → sampled from categorical distribution
- continuous actions: parameters of a probability distribution (one output node for each parameter, e.g., mean and standard deviation of Gaussian distribution) to sample actions from (and calculate log of the PDF for the sampled action to compute the policy gradient loss)

state → neural net → μ, σ → sample → continuous action

$\pi_w = \mathcal{N}\big(\mu_{\boldsymbol{w}}(s), \sigma_{\boldsymbol{w}}(s)\big)$ Gaussian used in A2C, PPO, SAC

# Problem: High-Variance Gradient Estimates

REINFORCE update uses full return $G_t$ (sum of future rewards) for all actions in the trajectory

→ noisy, unstable updates

- slow or no learning in complex environments
- sensitivity to reward scale and episode length

idea: reduce variance by subtracting a reference value (baseline)

then agent learns: “Was this action better or worse than expected?”

# REINFORCE with Baseline

again: $\mathbb{E}_{a\sim\pi}[\nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})] = 0$

expectation unchanged when subtracting an action-independent baseline:

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \mathbb{E}_{\pi}\left[\sum_t \nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w}) \left(G_t - b(s_t)\right)\right]$$

reduces variance if baseline is correlated with return:

if $b(s_t)$ close to $G_t$ → smaller variance in $G_t - b(s_t)$ → smaller variance in gradient

example for baseline: average discounted return

# Learned Value Function as Baseline

choose an estimate of the true state-value function as baseline:

$$b(s_t) = v(s_t|\boldsymbol{w}_v)$$

$$\boldsymbol{w} \leftarrow \boldsymbol{w} + \eta \cdot \sum_t \nabla_{\boldsymbol{w}_\pi} \log \pi(a_t|s_t; \boldsymbol{w}_\pi) \left(G_t - v(s_t|\boldsymbol{w}_v)\right)$$

e.g., separate networks
trained using MC returns

# Actor-Critic Methods

# Actor-Critic

TD error

use bootstrapping:

$$G_t - v(s_t|\boldsymbol{w_v}) \approx r_t + \gamma v(s_{t+1}|\boldsymbol{w_v}) - v(s_t|\boldsymbol{w_v}) = \delta_t$$

→ $v(s_t|\boldsymbol{w_v})$ is trained via TD learning

$\pi(a_t|s_t;\boldsymbol{w}_\pi)$: actor
$v(s_t|\boldsymbol{w_v})$: critic

update of $\pi(a_t|s_t;\boldsymbol{w}_\pi)$:

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \nabla_{\boldsymbol{w}_\pi} \log \pi(a_t|s_t;\boldsymbol{w}_\pi)\, \delta_t$$

→ policy updates can happen every time step

actor-critic = REINFORCE + learned baseline + bootstrapping
→ turning MC into TD method

# State-Action Form of Policy Gradient Theorem

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \mathbb{E}_{\pi}\left[\sum_{t} \nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})\, G_t\right]$$

$$= \mathbb{E}_{s,a\sim\pi}[\nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})\, Q_{\pi}(s_t,a_t)]$$

using
$Q_{\pi}(s_t,a_t) = \mathbb{E}_{\pi}[G_t|s_t,a_t]$

→ actions are weighted by their expected returns

theoretical foundation: exact gradient identity that actor-critic methods approximate

# Advantage Actor-Critic

advantage function: $A(s_t, a_t) = Q(s_t, a_t) - v(s_t)$

idea: calculates extra reward for specific action compared to average action in given state (how much better than expected)

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \mathbb{E}[\nabla_{\boldsymbol{w}} \log \pi(a_t | s_t; \boldsymbol{w})\, A(s_t, a_t)]$$

expectation unchanged when subtracting an action-independent baseline

# Advantage Actor-Critic (A2C / A3C)

advantage estimate: $A(s_t, a_t) \approx \delta_t$

→ same gradients as before (but different interpretation as advantage)

actor update: $\boldsymbol{w}_\pi \leftarrow \boldsymbol{w}_\pi + \eta \cdot \delta_t \nabla_{\boldsymbol{w}_\pi} \log \pi(a_t | s_t; \boldsymbol{w}_\pi)$

critic update: $\boldsymbol{w}_v \leftarrow \boldsymbol{w}_v + \eta_v \cdot \delta_t \nabla_{\boldsymbol{w}_v} v(s_t | \boldsymbol{w}_v)$

$$\text{target} = r_t + \gamma v(s_{t+1} | \boldsymbol{w}_v)$$

$$L(\boldsymbol{w}_v) = \frac{1}{2}\Big(v(s_t | \boldsymbol{w}_v) - \big(r_t + \gamma v(s_{t+1} | \boldsymbol{w}_v)\big)\Big)^2 = \frac{1}{2}{\delta_t}^2$$

$$\delta_t = r_t + \gamma v(s_{t+1} | \boldsymbol{w}_v) - v(s_t | \boldsymbol{w}_v)$$

$$\nabla_{\boldsymbol{w}_v} \delta_t \approx -\nabla_{\boldsymbol{w}_v} v(s_t | \boldsymbol{w}_v)$$

(semi-gradient method: treat target as constant)

# A2C vs A3C

A2C (Advantage Actor-Critic)

- synchronous updates
- uses batch of parallel environments
- more stable, GPU-friendly

A3C (Asynchronous Advantage Actor-Critic)

- multiple workers update independently
- no locking → faster but noisier
- CPU-friendly

- on-policy
- low variance (compared to REINFORCE)
- step-wise updates (fast learning)
- still sample inefficient
- can be unstable with function approximation

# Proximal Policy Optimization (PPO)

stable and efficient policy gradient (actor-critic) method

main problem with vanilla policy gradient (and also A2C): large updates → policy collapse, unstable training

PPO ≈ A2C
+ (better advantage estimation)
+ (safer policy updates)
+ (multiple epochs on same data)

# Generalized Advantage Estimation (GAE)

PPO uses GAE: a discounted sum of TD errors that interpolates between TD and Monte Carlo

A2C: 1-step advantage estimate: $A_t \approx \delta_t$

multi-step advantage estimate: $A_t^{(k)} \approx \delta_t + \gamma\delta_{t+1} + \cdots + \gamma^{k-1}\delta_{t+k-1}$

- larger $k$ → closer to Monte Carlo
- smaller $k$ → more bootstrapping

GAE combines all these with exponential weighting:

$$A_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty}(\gamma\lambda)^l\delta_{t+l}$$

equivalent form with $\lambda$-return $\hat{G}_t^{(\lambda)}$ : $A_t = \hat{G}_t^{(\lambda)} - V(s_t)$

$\lambda = 0$: pure TD, $\lambda = 1$: MC, PPO typically $\lambda \approx 0.95$

→ lower variance, smoother learning

advantages usually normalized to improve stability:

$$A_t \leftarrow \frac{A_t - \mathrm{mean}(a)}{\mathrm{std}(A)}$$

# Surrogate Objective

vanilla policy gradient (and A2C):

$$\nabla_{\boldsymbol{w}} J(\boldsymbol{w}) = \mathbb{E}_{\pi_{\boldsymbol{w}}}[\nabla_{\boldsymbol{w}} \log \pi(a_t|s_t;\boldsymbol{w})\, A(s_t, a_t)]$$

proxy objective (standard policy gradient surrogate):

$$L_{\mathrm{PG}} = \mathbb{E}_{\pi_{\mathrm{old}}}[r_t(\boldsymbol{w}_\pi) A(s_t, a_t)]$$

data from old policy

its gradient matches the true policy gradient at the current policy

using importance sampling to correct for mismatch (importance ratio $r_t$):

$$r_t(\boldsymbol{w}_\pi) = \frac{\pi(a_t|s_t;\boldsymbol{w}_\pi)}{\pi_{\mathrm{old}}(a_t|s_t;\boldsymbol{w}_{\mathrm{old}})}$$

likelihood ratio

problem: $r_t(\boldsymbol{w}_\pi) A(s_t, a_t)$ can become very large → unstable updates

# Clipping

key idea of PPO: constrained updates by means of clipped surrogate objective

clipping: instead of changing the policy arbitrarily, PPO restricts the update to keep the new policy "proximal" (close) to the old one

$$L_{\mathrm{PPO}} = \mathbb{E}_{\pi_{\mathrm{old}}}\left[\min\left(r_t(\boldsymbol{w}_\pi)A(s_t, a_t), \mathrm{clip}(r_t(\boldsymbol{w}_\pi), 1-\epsilon, 1+\epsilon)\, A(s_t, a_t)\right)\right]$$

- if update is small: normal policy gradient
- if update is large: clipped → prevents destructive updates

# Data Reuse

on-policy: collect data from $\pi_{\text{old}}$

A2C: use data only once (1 optimization epoch), then it's outdated
policy changes → gradients become incorrect → training becomes unstable
→ extremely sample-inefficient

PPO: data reuse (multiple optimization epochs)
possible because of clipped surrogate objective (controls policy drift)

# PPO Algorithm

1. collect trajectories with $\pi_{\text{old}}$
2. compute advantages (often with GAE)
3. optimize clipped objective for several epochs
4. update policy

- stable training, simple to implement
- works well with neural networks
- strong empirical performance

- still on-policy (with limited off-policy optimization) → sample inefficient
- needs careful tuning (clip range, epochs)

# Soft Actor-Critic (SAC)

in PPO, an entropy bonus can be added to the objective to encourage exploration

entropy: $\mathcal{H}(\pi) = -\mathbb{E}[\log \pi(a|s)]$

in SAC, entropy is part of the objective itself (maximum entropy actor-critic):

$$J(\pi) = \mathbb{E}\left[\sum_t r_t + \alpha\, \mathcal{H}\big(\pi(\cdot\,|s_t)\big)\right]$$

→ maximize reward and entropy

modified reward: $r_{\text{soft}} = r_t - \alpha \log \pi(a_t|s_t)$

# SAC: Critic and Actor

in training updates:
expectation over states → approximated by replay buffer $D$
expectation over actions → sampled from current policy $\pi_{\boldsymbol{w}}$

usually min of two critics
(reduces overestimation)

critic → soft Q-function(s):

soft Q-function: $Q^{\pi}(s,a) = \mathbb{E}_{\pi}[\sum_{k=0}^{\infty} \gamma^{k} \overbrace{(r_{t+k} - \alpha \log \pi(a_{t+k}|s_{t+k}))}^{r_{\text{soft}}}]$

learned with TD target (depending on actor policy)

data from replay buffer → off-policy

sampled from current policy

actor:

policy minimizes $L_{\text{SAC}}(\boldsymbol{w}) = \mathbb{E}_{s\sim D, a\sim \pi_{\boldsymbol{w}}}[\alpha \log \pi_{\boldsymbol{w}}(a|s) - Q^{\pi}(s,a)]$

→ move policy toward actions with high Q (critic)

with trade-off: maximize Q vs. maximize entropy

# Reparameterization Trick

the SAC actor objective $L_{\mathrm{SAC}}(\boldsymbol{w})$ involves the critic directly

→ to improve the actor: $\nabla_{\boldsymbol{w}} Q^{\pi}(s,a) = \frac{\partial Q^{\pi}}{\partial a}\frac{\partial a}{\partial \boldsymbol{w}}$

→ $a$ needs to be a differentiable function of $\boldsymbol{w}$

issue: actions are sampled from the policy, e.g., $a \sim \pi_w = \mathcal{N}\big(\mu_{\boldsymbol{w}}(s), \sigma_{\boldsymbol{w}}(s)\big)$, and sampling is non-differentiable in the usual sense

not necessarily Gaussian, just the most convenient choice

idea: rewrite it as $a = \mu_{\boldsymbol{w}}(s) + \sigma_{\boldsymbol{w}}(s) \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1)$

→ randomness is isolated in $\varepsilon$ (independent noise), everything else is deterministic and differentiable

since discrete actions are not reparameterizable, SAC only works for continuous actions

# SAC: Properties

sample efficient (off-policy)

strong exploration (entropy)

works well in continuous control

action bounding:
- A2C/PPO: simple clipping
- SAC: tanh squashing $a = \tanh(\mu + \sigma \cdot \varepsilon)$

but more complex than PPO:

- relies on learned soft Q-function(s) and off-policy bootstrapping
- coupled system: actor depends on critic, critic depends on actor, both change continuously
- more hyperparameters

AlphaGo, AlphaGo Zero

# Famous Example of Deep RL: AlphaGo

Monte Carlo tree search (heuristic, lookahead search) for move (i.e., action) selection

→ decision-time planning (advantage: focus on current state rather than full state space)

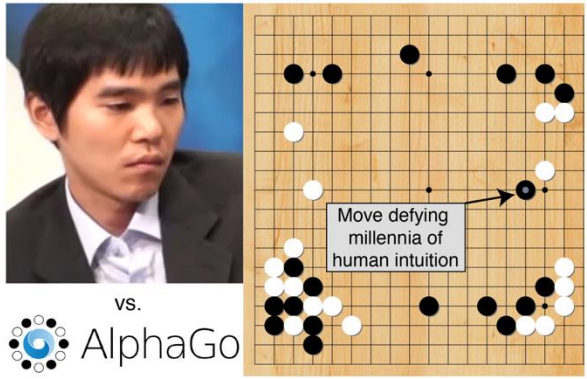
guided by deep convolutional neural networks for both value function and policy estimation

→ improving search efficiency

**reduce depth** of search tree by evaluating positions with **value function** (predicting outcome from given position → **bootstrapping**)

**reduce breath** of search tree by **sampling** actions using **policy network** (probability distribution over possible moves in given position)

famous move 37:




from Max Tegmark